

Introduction to Software Engineering and Models

- Foundational Concepts
 - The evolving role of software
 - Changing nature of software
 - Software myths
 - Software Engineering Principles
 - Software engineering as a layered technology
 - A process framework
 - Process patterns
 - Process assessment
 - Personal and team access models
 - Specific Process Models
 - The waterfall model
 - Incremental process models
 - Evolutionary process models
 - The unified process
-

Software

A software is basically a collection of programs, instructions, and data that tells a computer or electronic device how to perform specific tasks.

The Evolving Role of Software

- This refers to how software has transformed from simple programs in the mid-20th century (like basic calculators or payroll systems) to the backbone of the modern society.
- As software role's expands, it impacts economies, healthcare, entertainment, and more. For instance, during the Covid-19 pandemic, software allowed people to work remotely via tools like Zoom, Google Meet, etc. Understanding this evolution helps engineers anticipate future needs, like ethical AI, sustainable computing, etc.
- What it means for developers is that they must adapt to the new technologies by learning new paradigms, like shifting from desktop software to web-based services.
- **Example:** When Instagram first launched, it was just a simple photo sharing app but over the years it has evolved into a social commerce platform.